

Mary's Room: Can You Know Everything About Color and Still Not Know What Red Looks Like?

An original lesson plan inspired by **Martine Nida-Rümelin & Donnchadh O Conaill**'s the Stanford Encyclopedia of Philosophy entry on *Qualia: The Knowledge Argument*. Not affiliated with or endorsed by Stanford Encyclopedia of Philosophy or Martine Nida-Rümelin & Donnchadh O Conaill. Explore the original course: <https://plato.stanford.edu/entries/qualia-knowledge/>

GRADE LEVEL

Grades 11-12 / Intro College

DURATION

55 minutes

TOPICS

Philosophy of Mind

“If Mary knows every physical fact there is about color vision but has spent her whole life in a black-and-white room, does she learn something genuinely new the first moment she sees red — and if so, what does that tell us about whether consciousness can be fully explained in physical terms?”

Learning Objectives

By the end of this lesson, students will be able to:

- State Frank Jackson's Mary's Room thought experiment accurately and explain what conclusion it is meant to support about physicalism.
- Define 'qualia' and explain why the knowledge argument targets specifically the qualitative, felt character of experience rather than behavior or brain function.
- Evaluate at least one major reply to the knowledge argument, such as the Ability Hypothesis or the Old-Fact/New-Guise reply, and assess whether it successfully defends physicalism.

Materials Needed

- Handout presenting the Mary's Room thought experiment in full, with a simple illustration of the black-and-white room and monitor
- A set of physical color swatches (red, blue, green paper or paint chips) for the opening activity
- Reply-cards handout: the Ability Hypothesis and the Old-Fact/New-Guise reply, each with space for student rebuttals
- Whiteboard for tracking the class's running vote on whether Mary learns something new

Lesson Procedure

Hook: Describe Red to Someone Who's Never Seen It

8 min

In pairs, have one student attempt to describe, using only words and no comparisons to other colors, what red looks like to a hypothetical person who has been totally colorblind since birth. Students will struggle. Use this immediately to set up the thought experiment: what if that person knew every physical fact about color there is to know?

Direct Instruction: The Thought Experiment

15 min

Present Frank Jackson's scenario carefully: Mary is a brilliant scientist who has lived her entire life in a black-and-white room and has learned complete physical knowledge of color vision, including every fact about wavelengths and the neurophysiology of human vision, entirely through black-and-white materials. One day she leaves the room and sees a red tomato for the first time, and Jackson asks whether she learns something new. Introduce 'qualia' as the term for the subjective, felt qualities of experience, and explain the argument's structure: if physicalism is true, complete physical knowledge should be complete knowledge, but Mary seems to learn something upon leaving the room, so complete physical knowledge is not complete knowledge.

Small-Group Work: Testing the Argument

15 min

Groups evaluate the argument's premises and generate at least one objection of their own before seeing any official replies, pushing them to reason independently about whether the case really shows what Jackson claims.

Reply-Card Jigsaw: Physicalist Responses

12 min

Introduce the Ability Hypothesis, which holds that Mary gains new abilities to recognize, imagine, and remember red rather than new propositional knowledge, and the Old-Fact/New-Guise reply, which holds that Mary already knew the physical fact but comes to know it under a new mode of presentation. Groups assigned one reply must state it and construct the strongest response a defender of the knowledge argument could give back.

Closure: Final Vote

5 min

Retake the vote from the hook: does Mary learn something new upon leaving the room, and if so, does that refute physicalism, or can it be explained away? Ask for a show of hands and a one-sentence justification from a student on each side.

Discussion Questions

- 1 Does Mary really learn a new fact when she sees red for the first time, or does she just gain a new ability without learning anything propositional?
- 2 If the Ability Hypothesis is correct, does that fully capture what seems to change for Mary, or does something still seem left out?
- 3 Could Mary's 'complete physical knowledge' ever really be complete if she has never had the relevant experience, or is the thought experiment asking us to imagine something that isn't actually possible?
- 4 What is the difference between knowing a fact under a new guise and learning an entirely new fact, and does that distinction rescue physicalism from Jackson's argument?

- 5 If the knowledge argument succeeds, what would follow for the possibility of building a machine that has genuine conscious experience rather than merely processing information about colors?

Assessment

Students write a short paper explaining Jackson's knowledge argument in their own words, then pick either the Ability Hypothesis or the Old-Fact/New-Guise reply and argue whether it successfully saves physicalism from the argument, addressing at least one weakness in the reply they choose. Grade for accurate handling of the thought experiment's logical structure and for direct engagement with the chosen reply, not for which side the student takes.

Extension Activity

Have students design their own Mary's-Room-style thought experiment for a different sense or experience, such as a person raised without ever tasting anything sweet or without ever hearing music, and write a short analysis of whether their version supports the same conclusion Jackson draws, or whether it reveals a disanalogy that weakens the original argument.

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